Ex No 9 : Massive MIMO Capacity vs SNR and Gain Using MATLAB

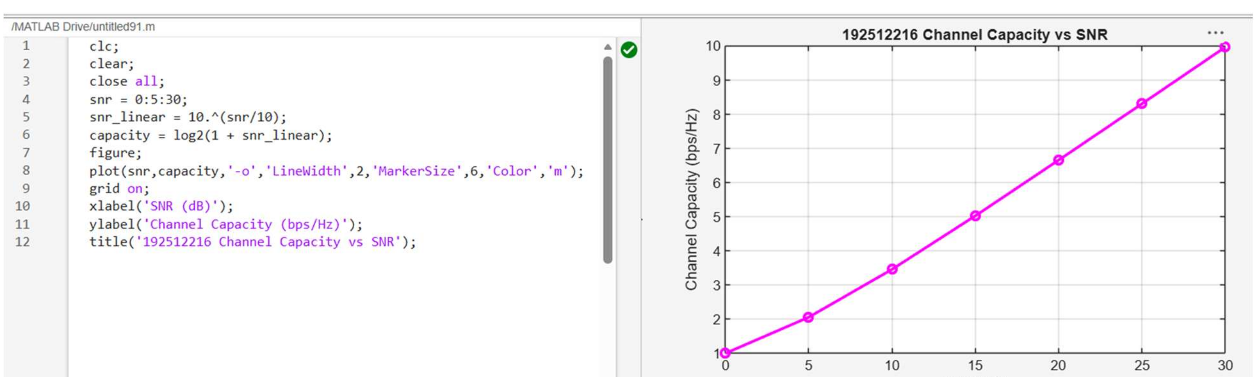
Aim

To analyze the performance of a Massive MIMO system by studying channel capacity, signal-to-noise ratio (SNR), and antenna gain using MATLAB.

Objective 1:

```
clc;

clear;

close all;

snr = 0:5:30;

snr_linear = 10.^(snr/10);

capacity = log2(1 + snr_linear);

figure;

plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);

grid on;

xlabel('SNR (dB)');

ylabel('Channel Capacity (bps/Hz)');

title('Channel Capacity vs SNR');
```

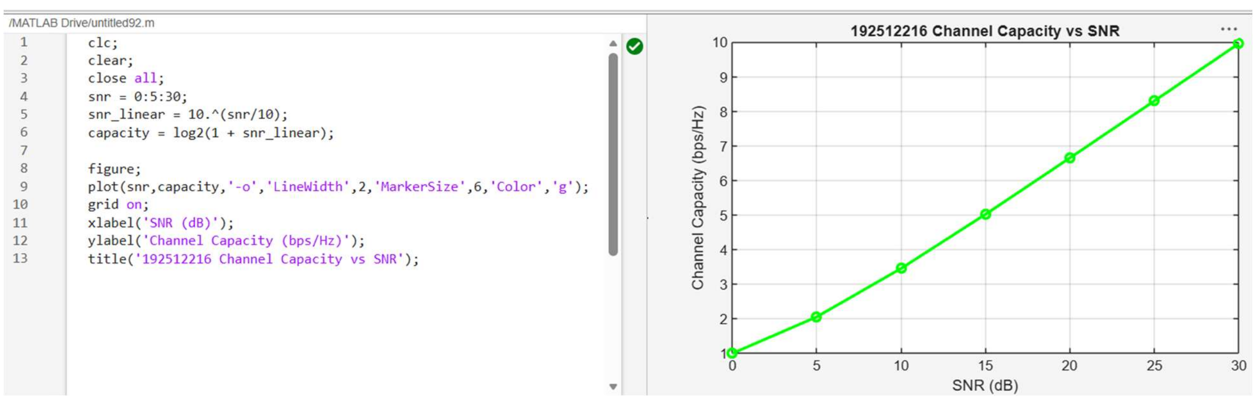
Output:



Objective 2:

```
clc;

clear;

close all;

snr = 0:5:30;

snr_linear = 10.^(snr/10);

capacity = log2(1 + snr_linear);


figure;

plot(snr,capacity,'-o','LineWidth',2,'MarkerSize',6);

grid on;

xlabel('SNR (dB)');

ylabel('Channel Capacity (bps/Hz)');

title('192512216 Channel Capacity vs SNR');
```

Output:



Objective 3:

```
clc; clear; close all;

antennas = [4 8 16 32 64];

gain = 10*log10(antennas);

figure
```

bar(antennas, gain)

grid on

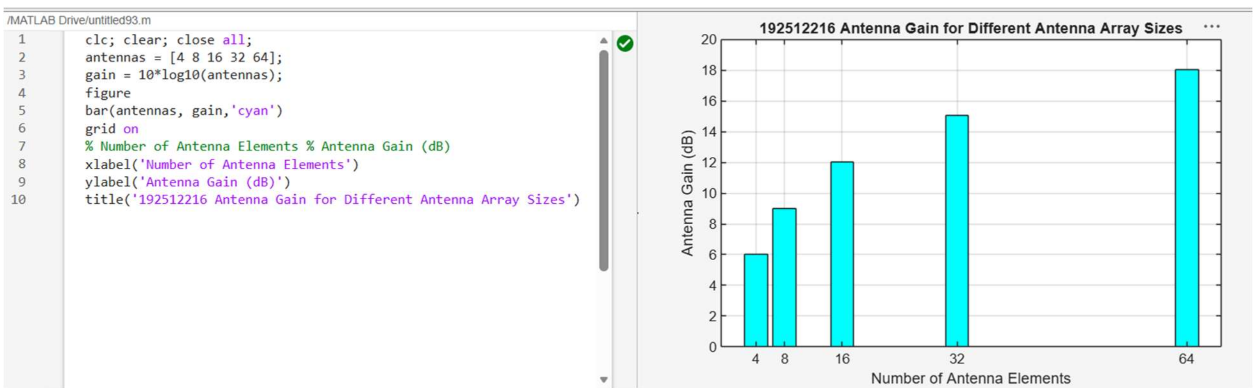
% Number of Antenna Elements % Antenna Gain (dB)

xlabel('Number of Antenna Elements')

ylabel('Antenna Gain (dB)')

title('Antenna Gain for Different Antenna Array Sizes')

Output:



Result:

The MATLAB analysis successfully demonstrated the relationship between channel capacity, signal-to-noise ratio, and antenna gain in a Massive MIMO system. The results showed that higher SNR and larger antenna arrays improve communication performance by increasing channel capacity and directional gain.